

Wireless Interfaces

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Course ID: CSE - 4619

Course Title: Peripherals, Interfacing and Embedded Systems

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Lecture References:

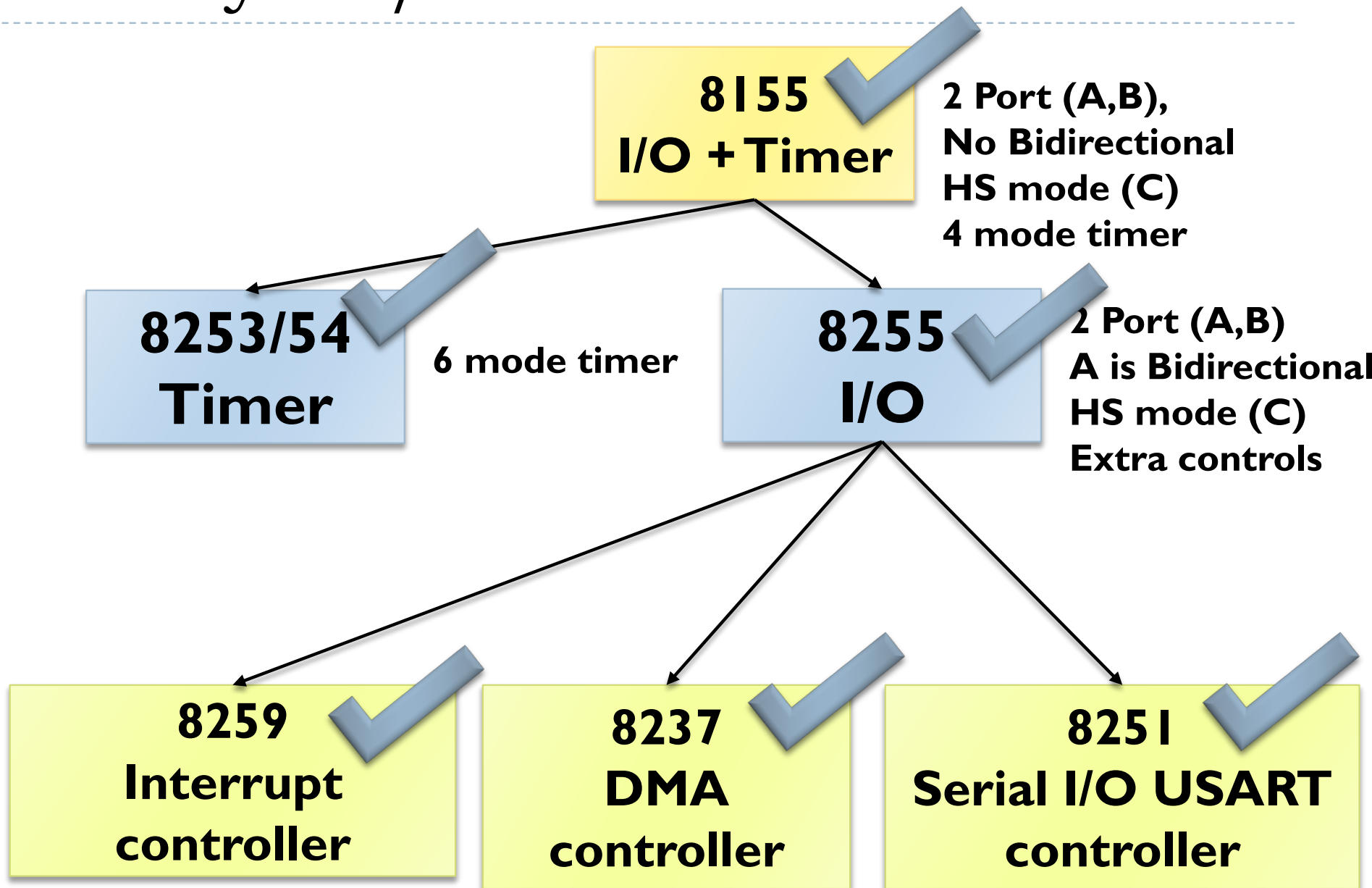
- ▶ **Book:**

- ▶ *Microprocessor Architecture, Programming and Applications with 8085 (Chapter-15)*, **Author:** Ramesh Gaonkor
- ▶ *Microprocessors and Interfacing: Programming and Hardware*, **Author:** Douglas V. Hall

- ▶ **Lecture Materials:**

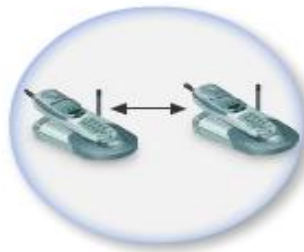
- ▶ *I/O System Design*, Dr. Esam Al_Qaralleh, CE Department, Princess Sumaya University for Technology.

Hierarchy of I/O Control Devices



History of Wireless Communication

- ▶ **Guglielmo Marconi invented the wireless telegraph in 1896**
 - ▶ Communication by encoding alphanumeric characters in analog signal
 - ▶ Sent telegraphic signals across the Atlantic Ocean
- ▶ **1914 – first voice communication over radio waves**
- ▶ **Communications satellites launched in 1960s**
- ▶ **Advances in wireless technology**
 - ▶ Radio, television, mobile telephone, communication satellites
- ▶ **More recently**
 - ▶ Satellite communications, wireless networking, cellular technology



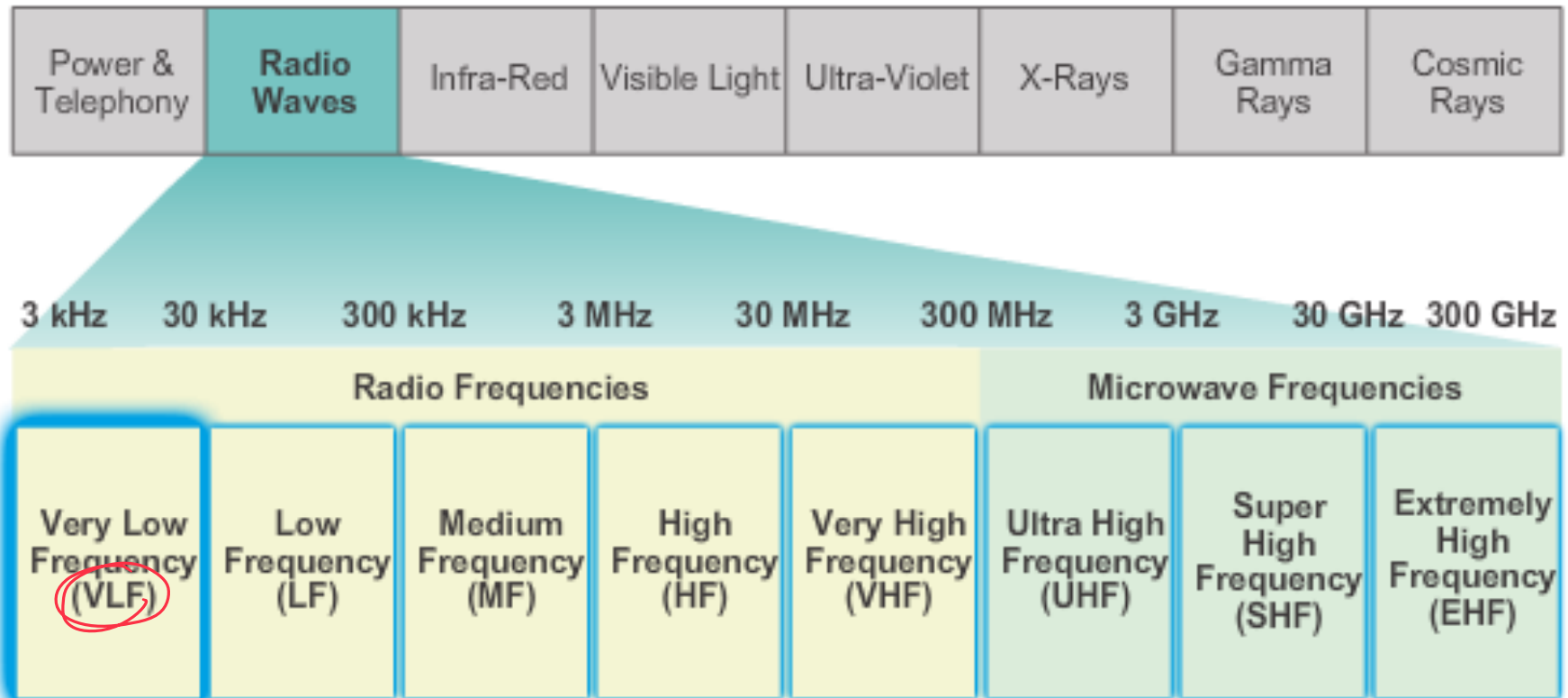
Wireless Frequency?

- ▶ Transmitting voice and data using electromagnetic waves in open space (atmosphere) or in any medium.
- ▶ Electromagnetic waves
 - ▶ Travel at speed of light ($c = 3 \times 10^8$ m/s)
 - ▶ Has a frequency (f) and wavelength (λ)
 - $c = f \times \lambda$
 - ▶ Higher frequency means higher energy photons
 - ▶ The higher the energy photon the more penetrating is the radiation

*IR: - relatively cheap
- needs line of sight
- diode emits infrared light to generate signals.
Infrared transistor catches this signal

Wireless Frequency

Radio Waves of the Electromagnetic Spectrum



Wireless Communication Supported by Very Low Frequency (VLF)

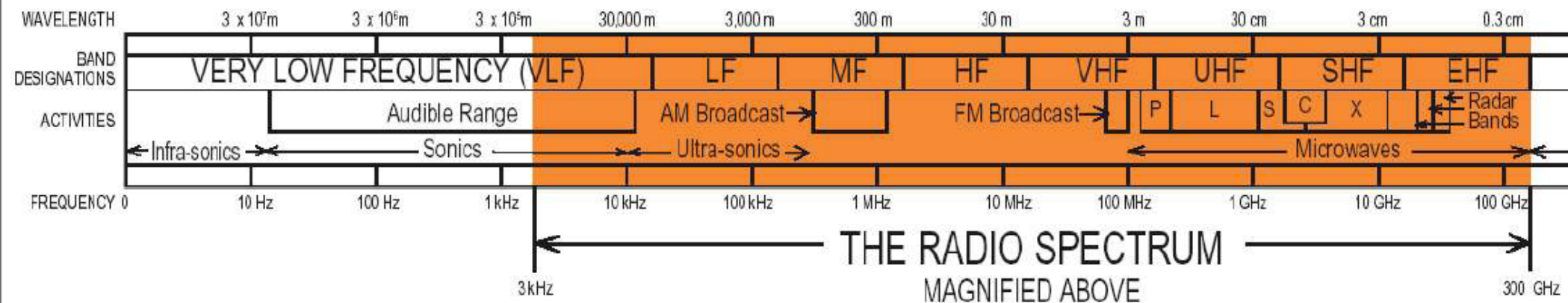
Radio navigation

Submarine communication

Wireless heart rate monitors

What is RF?

Radio Frequency is an electromagnetic signal with a frequency between **3 kHz** and **300 GHz**

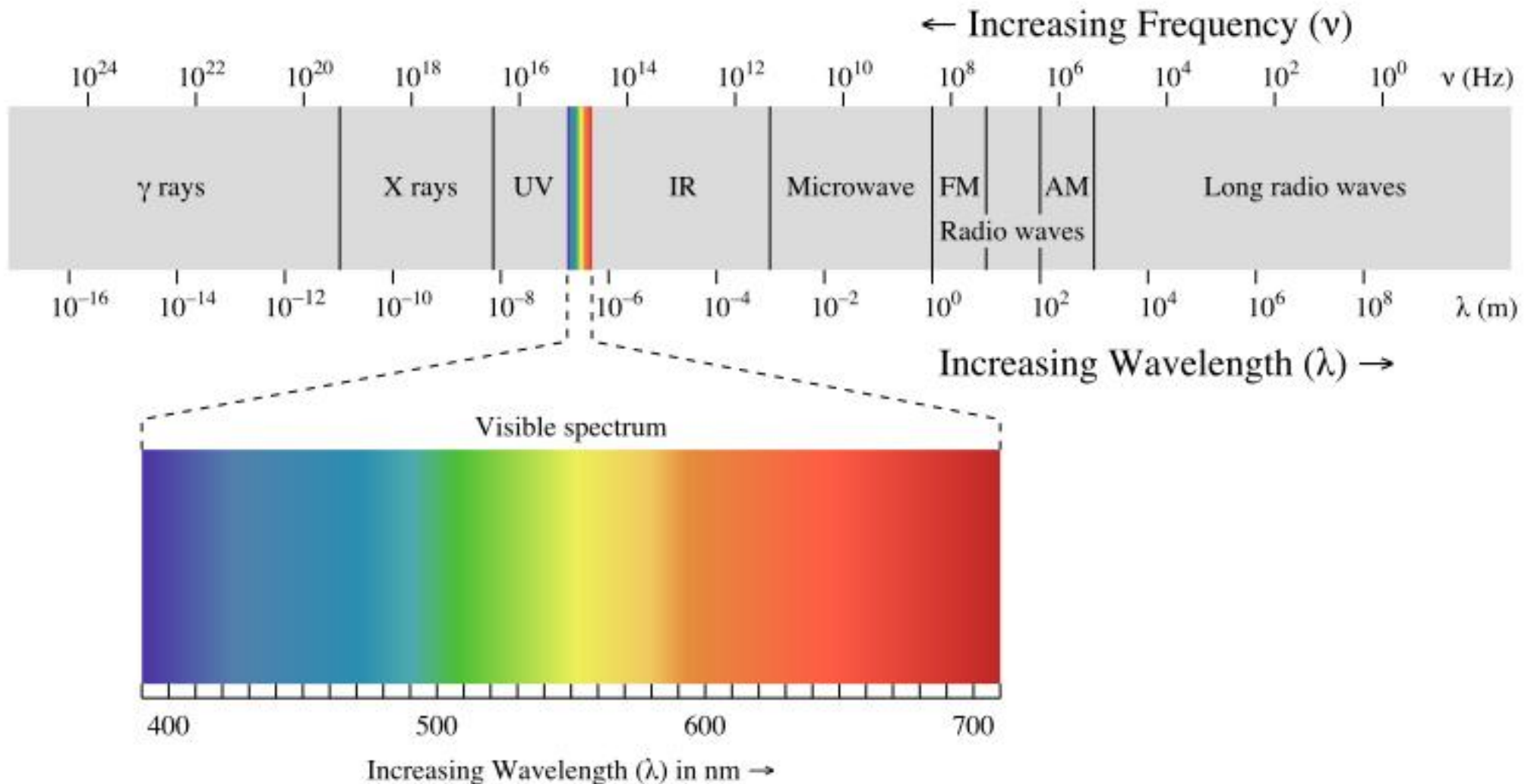


RF signals carry analog or digital information

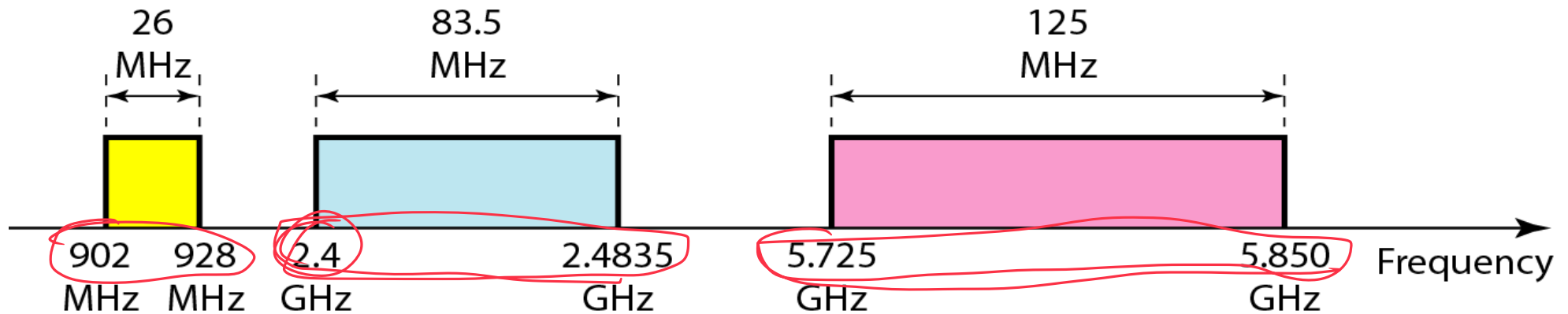
- Analog: Information content varies continuously over time
 - *Example: radio and TV stations*
- Digital: Information content consists of discrete units (e.g., 0s and 1s)
 - *Example: Cell phones and wireless networks*

- LOS not necessary
- longer distance communication
- frequency hopping

Electromagnetic Radiation Spectrum



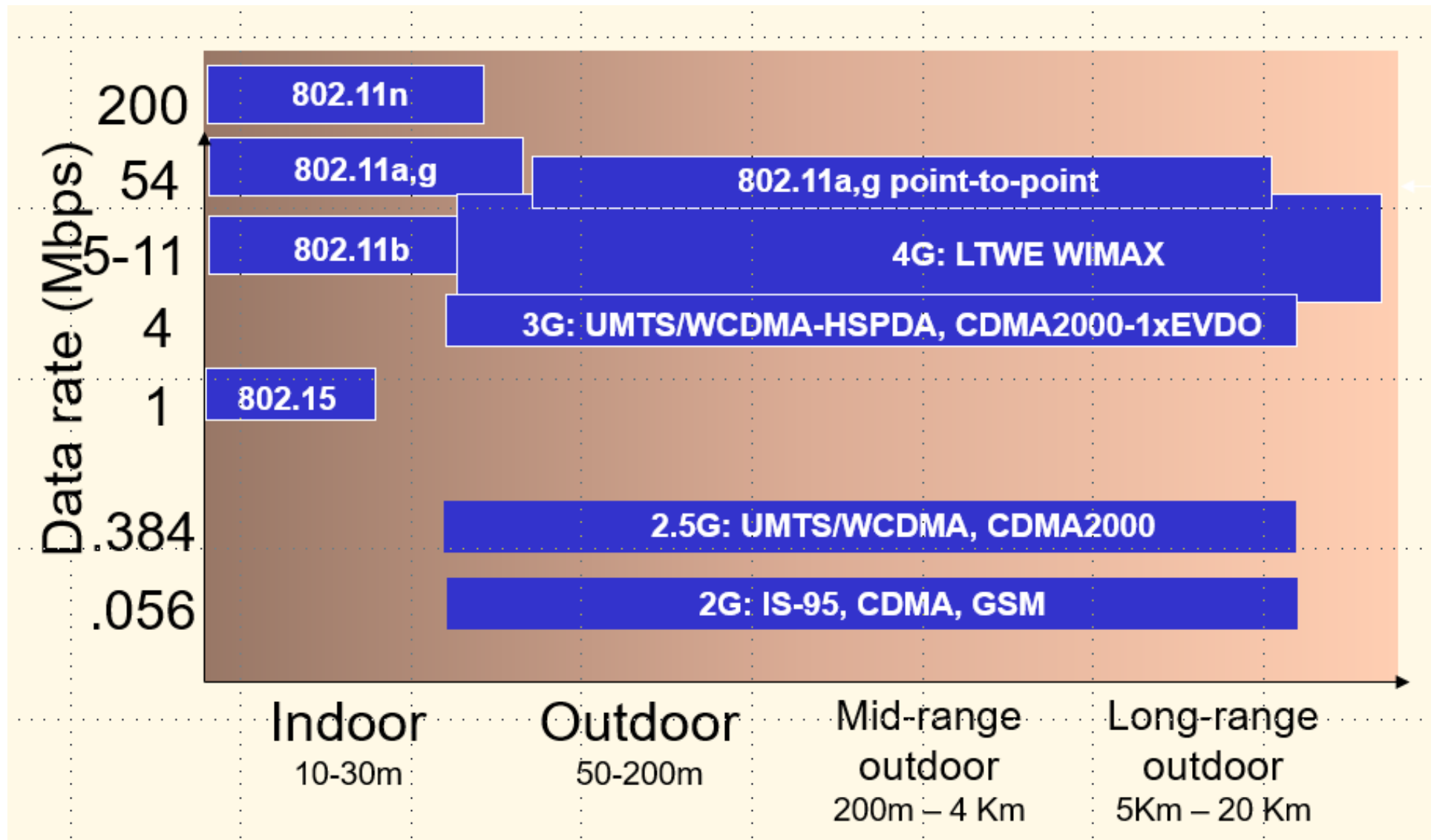
Industrial, Scientific & Medical (ISM) Band



Wireless Interfaces

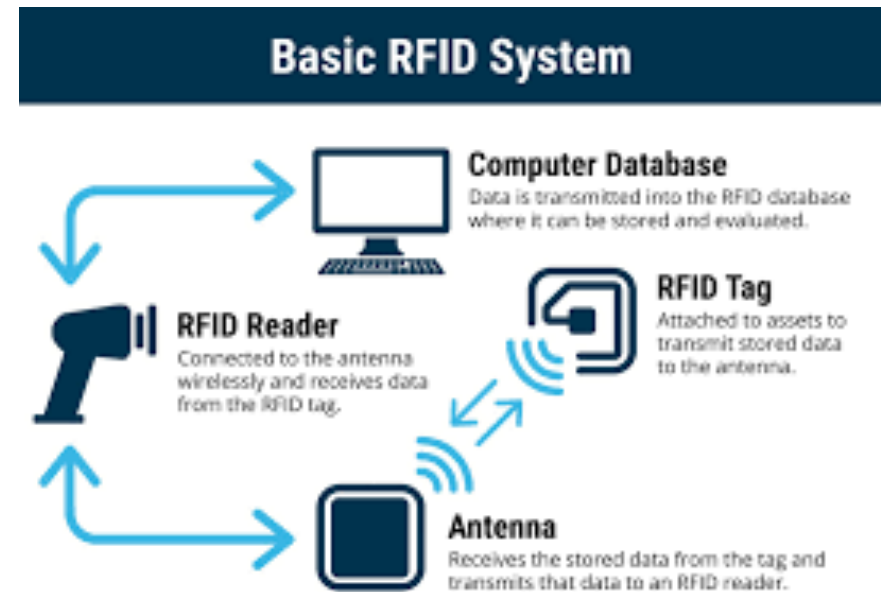
- ▶ RFID (Radio Frequency Identifier)
- ▶ BT (Bluetooth)
- ▶ WiFi (Wireless Fidelity)
- ▶ Cellular Technology
- ▶ LoRa IoT Interface Standards

Wireless Link Standards



RFID Interfaces

- ▶ RFID uses **radio waves** to transfer data from an electronic tag (RFID tag or label), attached to an object, through a reader to identify and track the object.
- ▶ The tag's information is stored electronically.
- ▶ Some RFID tags can be read from several meters away and beyond the line of sight of the reader.



RFID Interfaces

- ▶ An RFID reader transmits an encoded radio signal to interrogate the tag.
- ▶ With a small RF transmitter and receiver, the RFID tag receives the message and responds with its identification information.
- ▶ Many RFID tags have no battery. Instead, the tag uses the radio energy transmitted by the reader as its energy source.

Interfaces: Wireless Protocols

- ▶ **IrDA** (Infrared Data Association)
 - ▶ IrDA is an international organization
 - ▶ Designed to support transmission between two devices over short-range point-to-point infrared. *ie: tv to remote*
 - ▶ Rate: 9.6 Kb/s – 4 Mb/s
 - ▶ Deployed in notebooks, printers, PDA's, cell phones,...
 - ▶ MS Windows CE 1.0 the first Windows OS support it
 - ▶ Available on several popular embedded OSs

Bluetooth Interfaces

very widely used

- ▶ Bluetooth is a wireless technology which is used for computers, PDAs, mobile phones, headphones, mice and keyboards.
- ▶ It transmits very weak signals on a 2.4 to 2.48 gigahertz or 5 gigahertz frequency (ISM Band) for communication over short distances, typically up to 10 metres.

Bluetooth $\Rightarrow$ for exchanging I/O signals
uses RF (radio frequency) $\left\{ \begin{array}{l} \rightarrow \text{Radiowave (RW)} \\ \rightarrow \text{microwave (MW)} \end{array} \right.$

Bluetooth transfer speed up to 1 megabit per second.

ISM Band
 $\swarrow$ 900MHz
 $\downarrow$ 2.4-2.48GHz
 $\rightarrow$ 5.25-5.75GHz

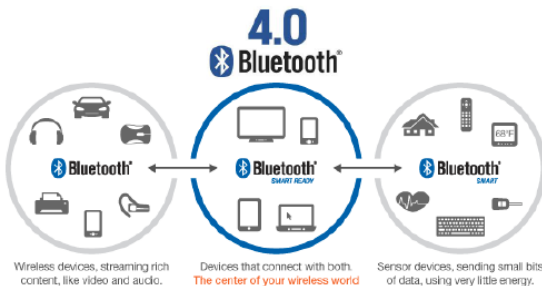
Bluetooth Interfaces

BT Basic/Enhanced Data Rate

- 2.45 GHz ISM frequency band
- 79 Channels (USA, Europe) or 23 (Japan) RF channels. 1 MHz channel width
- Frequency Hopping (FH) - Frequency-Hopping/Time-Division-Duplex (FH/TDD) - Adaptive Frequency Hopping
- Devices are organized in piconets
- The master of the piconet provides the clock and the frequency-hopping pattern (ordering of the frequencies)

BT Low Energy

- 2.45 GHz ISM band
- 40 RF channels. 2 MHz channel width.
- AFH mechanism FDMA and TDMA
- @FDMA: the 40 physical channels are divided into 3 advertising channels and 37 data channels.
- Devices are organized in piconets.
- The master of the piconet determines the hop interval and the hopping pattern for accessing the 37 physical channels.
- @TDMA: polling-based data transmission between the master and the slaves of the piconet.



WiFi Interfaces

IEEE 802.11 a/b/g/b/ac
↓
600 mbps

▶ IEEE 802.11

- ▶ IEEE proposed standard for WLAN
- ▶ Ad-hoc vs. infrastructure
- ▶ PHY and MAC layers
- ▶ Data rate: 1 Mbps, 2 Mbps, 11 Mbps, 54 Mbps, 600 Mbps
- ▶ Calls: 2.4 – 2.4835 GHz frequency band (unlicensed band).
- ▶ Use CSMA/CA
- ▶ Signals for transmission: RTS, CTS, and ACK.

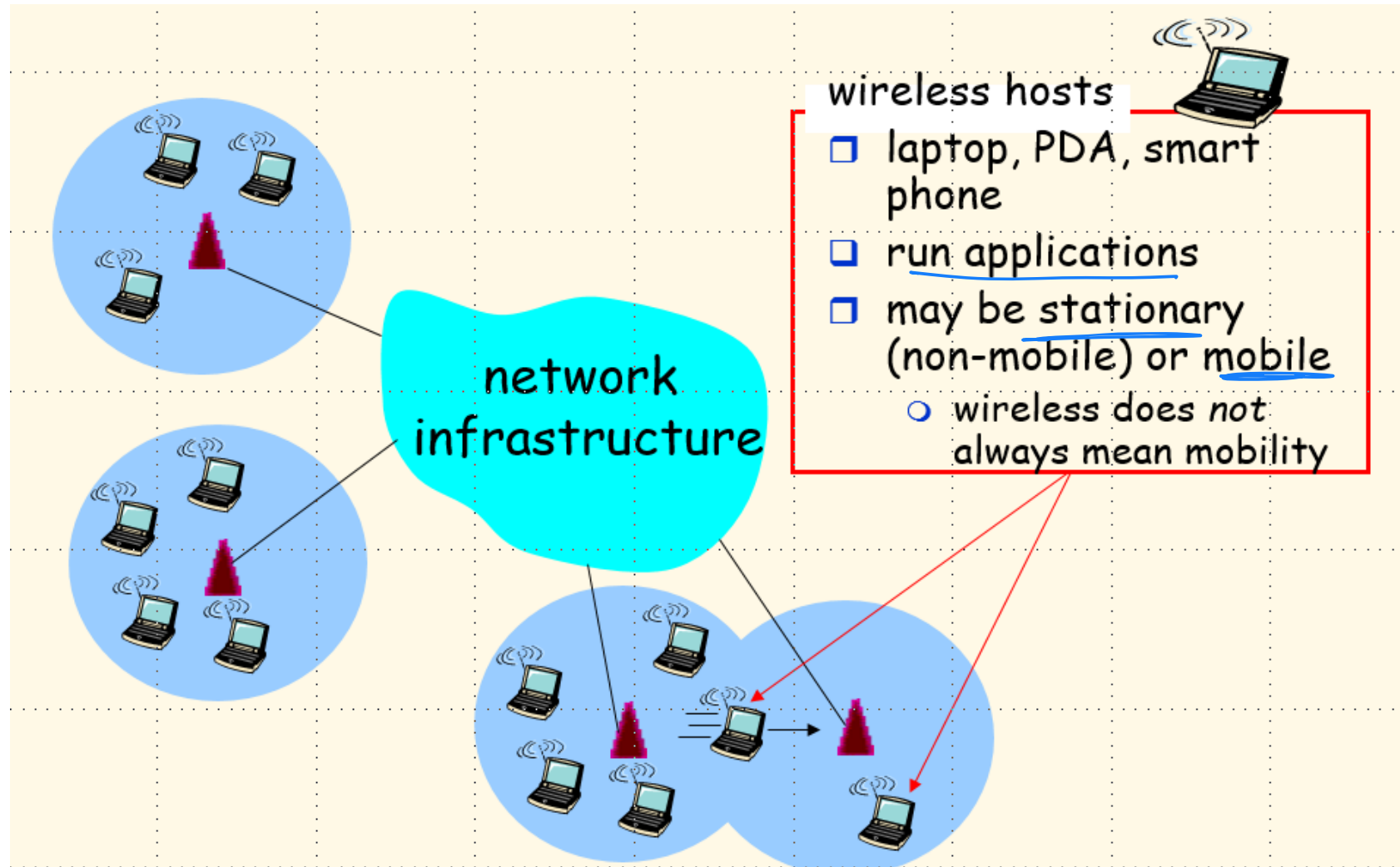


Wifi uses RF (radio frequency)

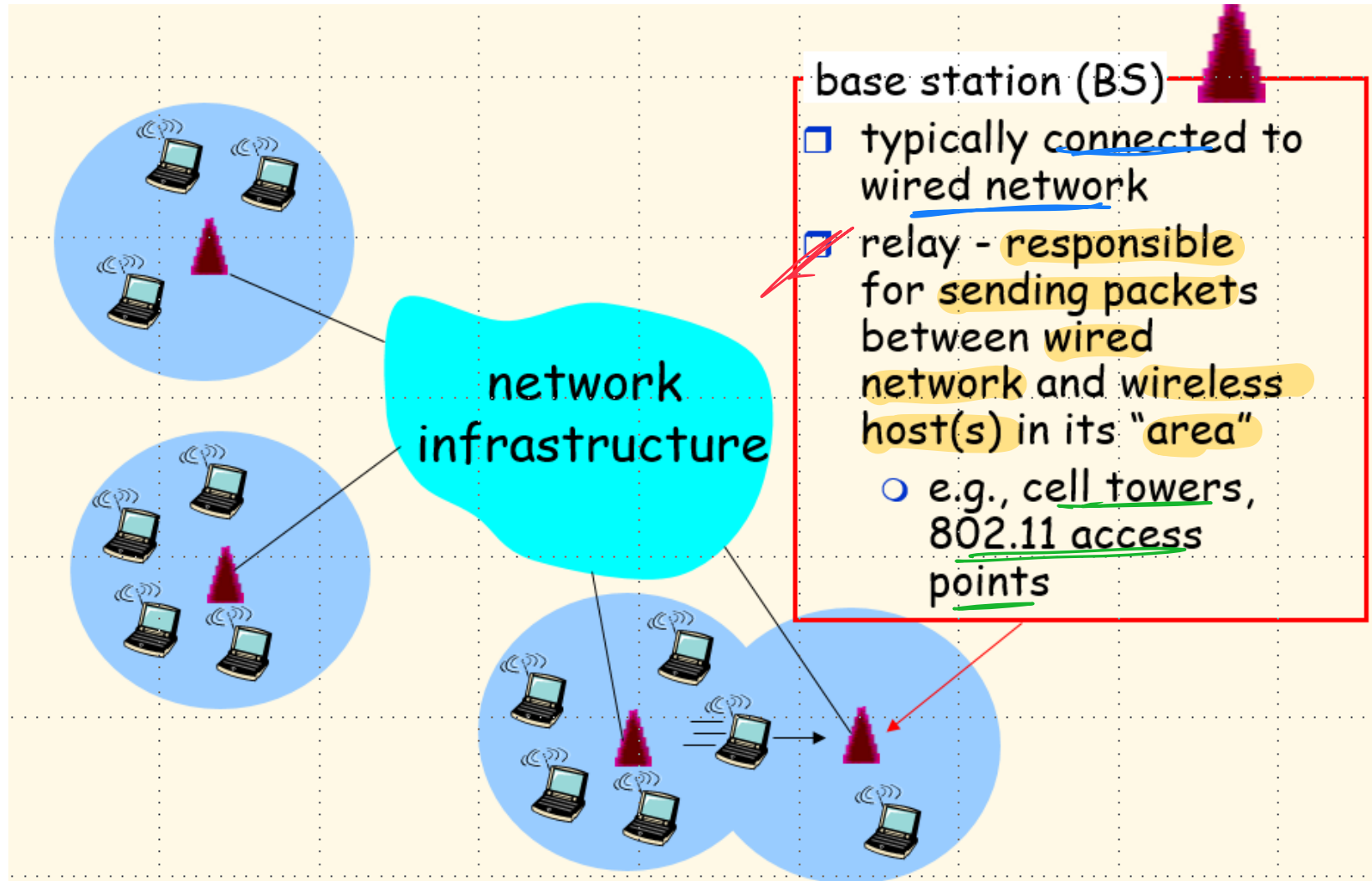
300 kHz ————— 300 GHz

Regulatory committee (BTRC) which sells these frequencies

WiFi Interfaces



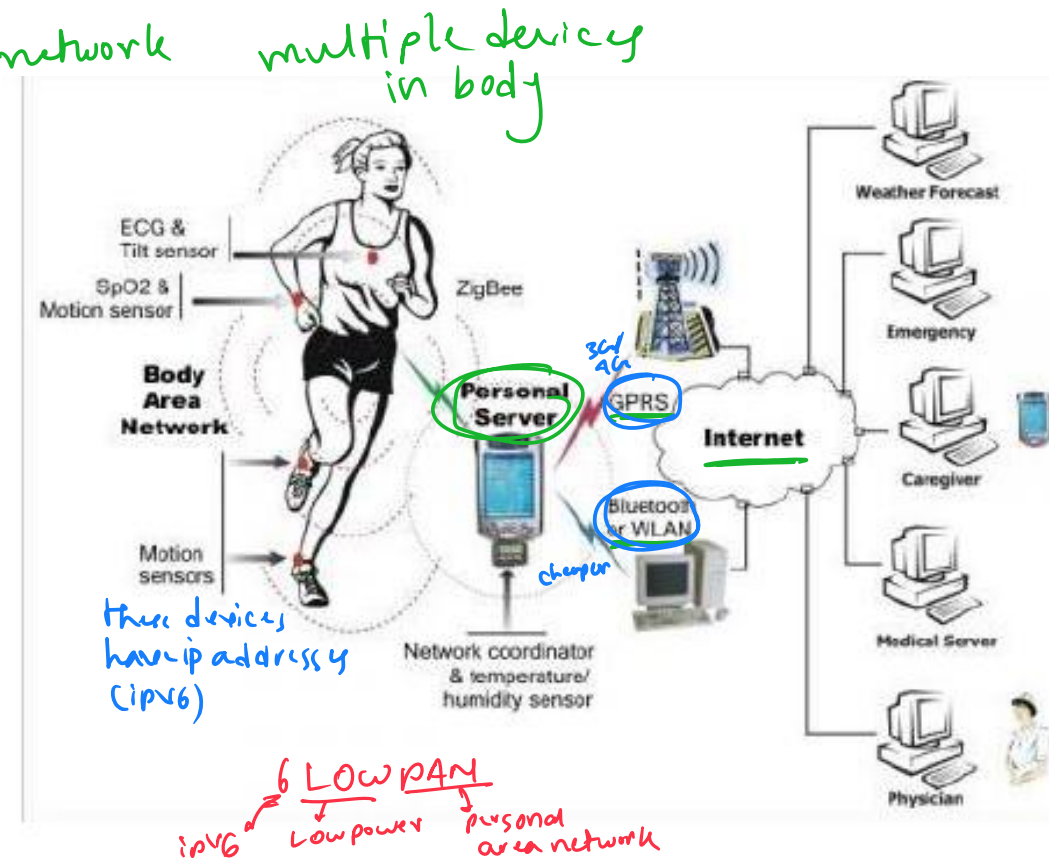
WiFi Interfaces



WPAN Interfaces

▶ IEEE 802.15.4

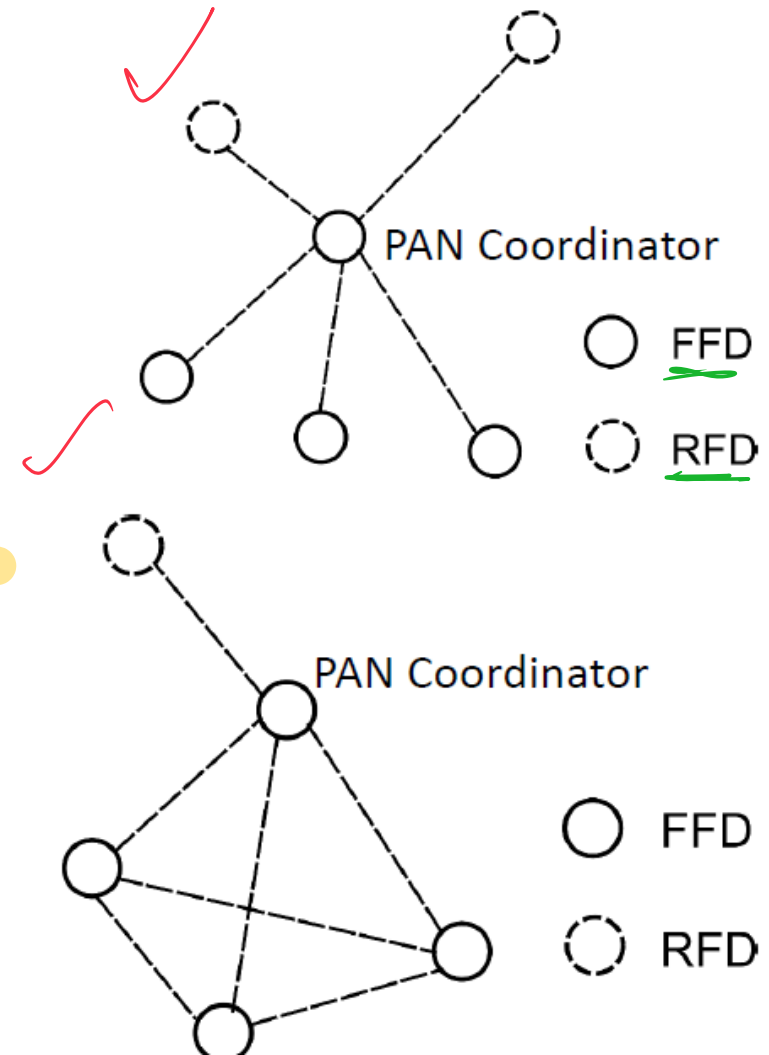
- ▶ IEEE proposed standard for **WPAN** *wireless personal area network*
- ▶ Basically for star-topology
- ▶ PHY and MAC layers
- ▶ Data rate: 250 Kbps
- ▶ Calls 2.4 – 2.4835 GHz frequency band (unlicensed band).
- ▶ Use CSMA/CA and TDMA



WPAN Interfaces

Types of logical topologies

- **Star:** each device (FFD or RFD) communicates with the PAN coordinator only
 - Suitable for small-scale networks that operate within a limited space
 - home automation, computer peripherals, peripherals, games, and smart health care
- **Peer-to-peer:** FFD devices can communicate with each other, as long as they are within communication range.
 - More flexible than star, suitable for larger-scale networks that need distributed coordination between peers without the necessity of a central unit.
 - Multi-hopping, Cluster Trees, and Mesh networking



Embedded IoT Network

↪ based on chip spread spectrum modulation

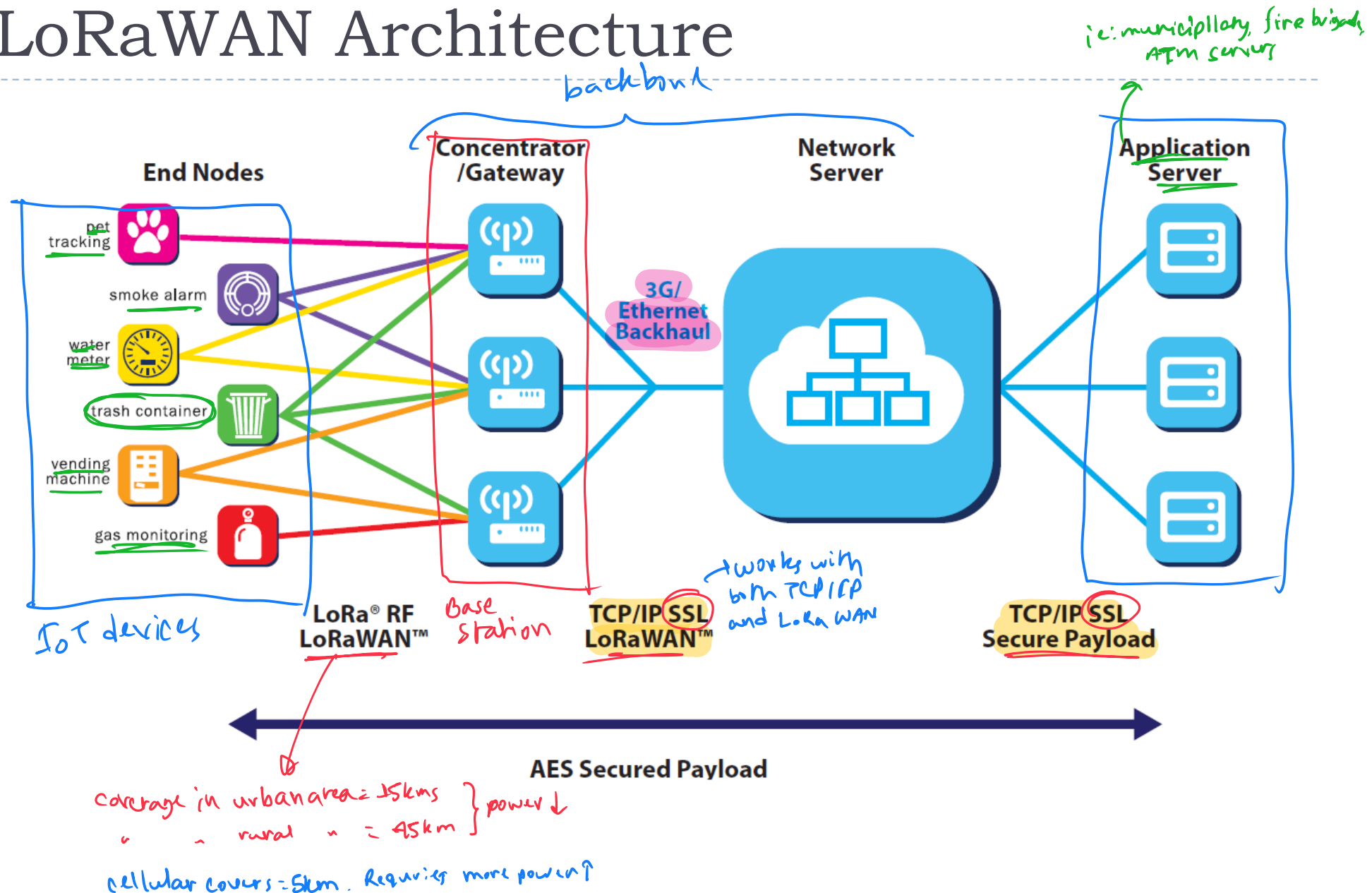
LoRaWAN: Long Range WAN

- ▶ The advantage of LoRa is in the technology's **long range capability**.
- ▶ A single gateway or base station can cover entire cities or hundreds of square kilometres.
- ▶ Range highly depends on the environment or obstructions in a given location, but LoRa and LoRaWAN have a **link budget greater than any other standardized communication technology**.
- ▶ The link budget, typically given in decibels (dB), is the primary factor in determining the range in a given environment.

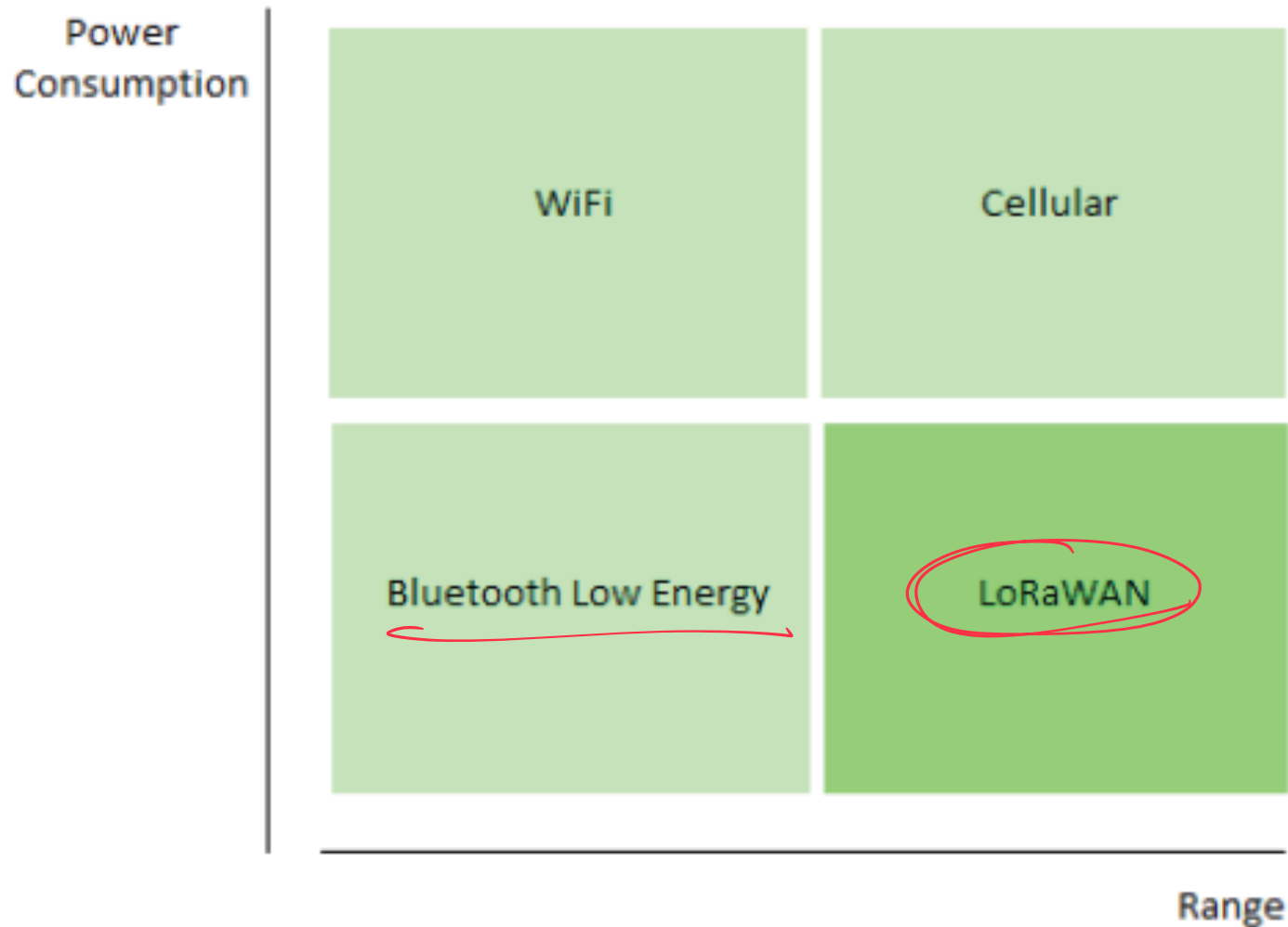
WHERE DOES LPWAN FIT?

- ▶ One technology cannot serve all of the projected applications and volumes for IoT.
- ▶ **WiFi and BT** are widely adopted standards and serve the applications related to **communicating personal devices** quite well.
- ▶ **Cellular technology** is a great fit for applications that need **high data throughput** and have a **power source**.
- ▶ **LPWAN** offers **multi-year battery lifetime** and is designed for **sensors and applications** that need to send **small amounts of data over long distances** a few times per hour from varying environments ✂ ✂

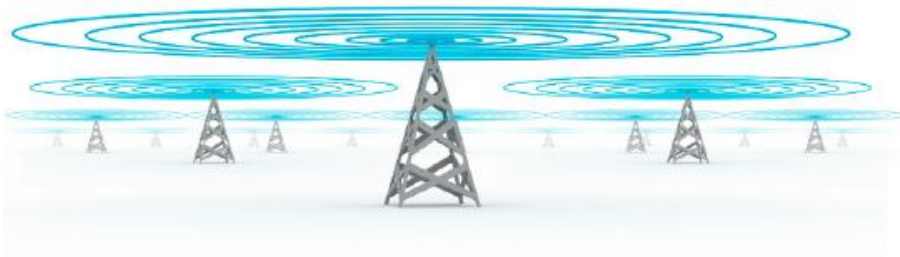
LoRaWAN Architecture



Wireless Technology Comparisons for IoT and Embedded Devices



LoRaWAN: Advantages



Long Range

- Greater than cellular
- Deep indoor coverage
- **Star topology**
 - one way vs two way communication



Max Lifetime

- Low power optimized
- 10-20yr lifetime
- **>10x vs cellular M2M**
 - 10 times higher

LoRaWAN: Advantages



Multi-Usage

- High capacity
- Multi-tenant
- Public network



Low Cost

- Minimal infrastructure
- Low cost end node
- Open SW
→ open source software

Thank You !!

